

*Epiphany is not about solving a complex puzzle, but
something that was too simple to see.*

Paper and Thesis

Keyao Peng. Sheaves and differential equations: An introduction to algebraic analysis, bachelor dissertation, 2019.

Keyao Peng. Milnor-witt motivic cohomology of complements of hyperplane arrangements. *Algebraic & Geometric Topology*, 23(8):3531–3552, 2023.

Keyao Peng. Milnor-witt motivic cohomology and linear algebraic groups. *Advances in Mathematics*, 458:109973, 2024.

Keyao Peng. Milnor-witt motivic decomposition of stiefel varieties. 2024.

Work Experience

2024–2027 **Post-doc**, *Institut de Mathématiques de Bourgogne(IMB)*, CNRS, Dijon, France

Education

2020–2023 **Ph.D.**, *Institut Fourier, Université Grenoble Alpes*, Grenoble, France
Algebraic geometry, with advisor *Jean Fasel*

2019–2020 **Master**, *Institut Fourier, Université Grenoble Alpes*, Grenoble, France
Mathématiques fondamentales

2015–2019 **Bachelor**, *Taishan College, Shandong University*, Jinan, China
Majored in mathematics

Academic Activities

Speaker

Nov 2024 Motivic homotopy in interaction *CIRM*

My talk: *Cellular A1-homology of toric surfaces*

2024 Séminaire Doctorant *IMB*

My talk: *$0=1-1=-1+1=0$: From Elementary School to Higher Algebras*

Aug 2023 Chow-Witt Rings: Computations and Applications *BIMSA*

My talk: *MW-motivic cohomology of linear algebraic groups and Stiefel varieties*

2022 Géométrie réelle, motifs et A1-homotopie *ENS de Lyon*

My talk: *Théorie des 6 foncteurs (Theory of six functors)*

2020–2021 Working group on stratified homotopy theory *IAS*

My talk: *Oriented pushouts and oriented fibre products*

2021-2023 Séminaire Compréhensible *Institut Fourier*

My talks:

May 2023 *How to explain (higher) categories to a geometric topologist? (An introduction to Cobordism Hypothesis)*

Feb 2021 *Homotopy type theory for mathematicians*

Teaching

Autumn 2022 **TA**, Université Grenoble Alpes, An introduction to algebraic geometry

Skill

Language

Chinese	Mother tongue	English	C1
French	B2	German	A1
Japanese	A2		

Computer

Language	Typescript, C#, Haskell, Purescript, Lean
Animation	Blender, Unity

Interest

Synthetic Geometry	Study geometry without using analysis, including algebraic geometry, arithmetic geometry, algebraic analysis and more
Homotopy	The geometry of "path", like A1 homotopy theory, homotopy type theory
Higher Structures	Higher means add homotopy to everything, like set, algebra, category, topos, TQFT, etc.